

Hydrostatic

Hydrostatic: Is the study of fluid at rest. fluids are substances that cannot withstand shear stress. They flow when external force is applied on them; they do not have a definite shape; they constantly collide with the walls of their container; they are incompressible and takes the shape of their container.

Hydrostatic has many application in engineering (for storing, transporting and using fluids) in geophysics and astrophysics (for understanding anomalies in the earth's gravitational fields); in medicine (for measuring blood pressure). etc.

Properties of fluids

- (1) Density: Is defined as the mass per unit volume. its unit is kg m^{-3}
- 2) Specific weight: Weight per unit volume. its unit is Nm^{-3} . It depend on the gravitational force.
- 3) Specific Volume: Mass per unit volume. its unit is $\text{m}^3 \text{ per kg}$.
- 4) Relative density (specific gravity): Ratio of density of substance to density of water at STP
- 5) Viscosity: Measure of the resistance to flow in fluid
- 6) Compressibility: Volume ~~de~~ decrease in fluid when pressure is applied.
- 7) Elasticity: The ability of fluids to return to their original volume when pressure (force) is applied.
- 8) Vapor pressure: Is the pressure at which the pressure of the surface of the fluid a liquid equal the pressure of liquid.
- 9) Surface tension: Is the force acting perpendicularly per unit length on surface of a liquid. Its unit is Nm^{-2}
- 10) Capillarity: The tendency of a liquid to rise and fall in a narrow tube.

Density and Relative density

Density (ρ) of a substance is defined as its mass per unit volume.

$$\rho = \frac{\text{mass (m)}}{\text{Volume (V)}} ; \text{kgm}^{-3} \quad \text{--- (1)}$$

Relationship between Weight and Density

$$\rho = \frac{m}{V} ; m = \rho V ; \text{Weight of a body } W = mg$$

$$\therefore W = \rho V g$$

for an ideal gas

$$PV = nRT ; n = \frac{\text{mass (m)}}{\text{molar mass (M)}}$$

$$PV = \frac{m}{M} RT ; PVM = m RT ; PM = \frac{m}{V} RT = \rho RT$$

$$\rho = \frac{PM}{RT}$$

Specific gravity (Relative density) ρ_r

This tells us how denser, or lighter, a given body is by comparing the mass, weight, or density of a given fluid with the same amount of water at 4°C

$$\begin{aligned} \rho_r &= \frac{\text{density of substance}}{\text{density of equal Vol. of water}} \\ &= \frac{\text{mass of substance}}{\text{mass of equal Vol. of water}} \\ &= \frac{\text{Weight of substance}}{\text{Weight of equal Vol. of water}} \end{aligned}$$

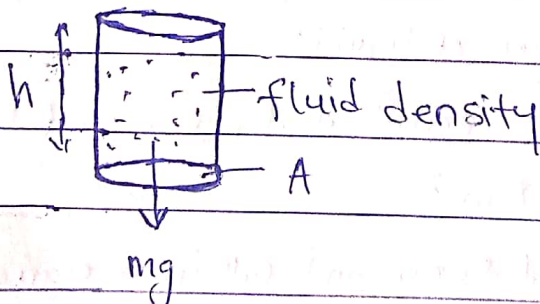
Pressure of a fluid

Pressure is the force acting perpendicularly per unit area

$$p = \frac{\text{force}}{\text{Area}} = \frac{F}{A} ; \text{Nm}^{-2} = 1 \text{ pascal (Pa)}$$

In a fluid, hydrostatic pressure is defined as $p = \rho g h$?

Consider a cylinder holding a fluid of density ρ at a depth h whose surface area is A as shown below.



One to the weight of the fluid; the hydrostatic pressure can be defined as;

$$p = \frac{F}{A} = \frac{mg}{A} = \frac{\rho V g}{A} = \frac{\rho A h g}{A} = \rho g h$$

$$p = \rho g h$$

Factors affecting hydrostatic pressure! For a given fluid, the pressure depends only on depth density and it acts equally in all direction. If the liquid is exposed to the atmosphere; the total pressure at a certain level above the liquid is $P = P_0 + \rho gh$.

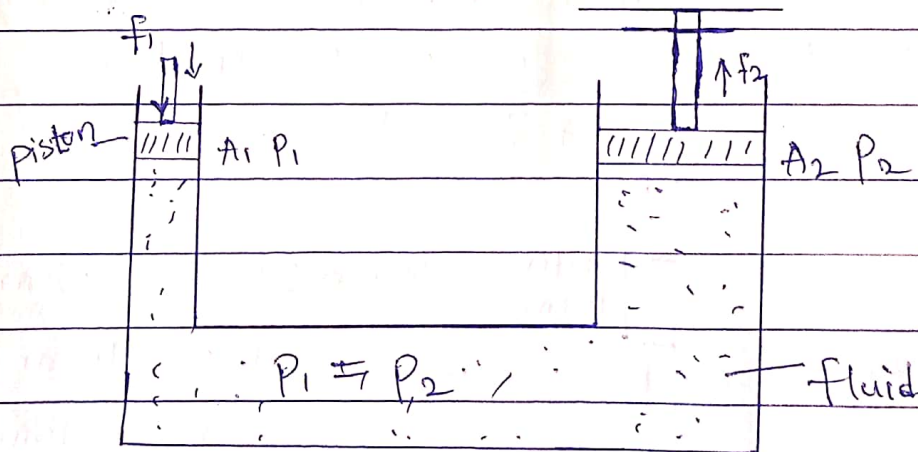
Pascal Principle: This states that for an incompressible fluids in a closed container, the pressure exerted on one surface of the fluid is equal to the pressure on any other surface

or

for a fluid at rest in a closed container, a pressure change in one part is transmitted without loss to every portion of the fluid and to the walls of the container.

Application of Pascal principle

The principle allows a small force applied to a small surface area to produce a large force which is used in the hydraulic systems as shown below:



Used as hydraulic lift/press

According to Pascal's principle $P_1 = P_2$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

$$F_2 = \frac{F_1 A_2}{A_1}$$

Example :

A Car is raised by applying a force of 150 N to a hydraulic piston of area 0.01 m^2 . If the car rests on a piston area of 0.1 m^2 , what will be the mass of the car that the piston can raise? Assume the piston mass is 5 kg .

Solution $F_1 = 150$; $A = 0.01\text{ m}^2$; $A_2 = 0.1\text{ m}^2$

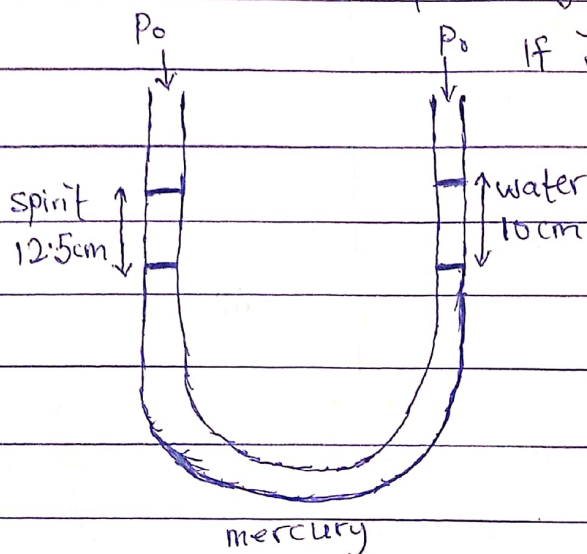
$$F_2 = \frac{F_1 A_2}{A_1} \quad (\text{Pascal's principle})$$

$$= \frac{0.1 \times 150}{0.01} = 1500\text{ N}. \text{ This is the maximum force}$$

Used to lift the Car. This force is balanced by the force of gravity given as; $F = mg$ or $m = \frac{F}{g} = \frac{1500}{9.8} = 153\text{ kg}$

Since the piston has a mass of 5 kg ; maximum mass of the car is $153\text{ kg} - 5\text{ kg} = 148\text{ kg}$

Question: A U-tube contains water and methylated spirit separated by mercury. The mercury column in the two arms are in level with 10.0 cm of water in one arm and 12.5 cm of spirit in the other. What is the specific gravity of the spirit?



If $\rho_w = 1\text{ g cm}^{-3}$

Given: $h_s = 12.5\text{ cm}$

$$h_w = 10\text{ cm}$$

$$\rho_w = 1\text{ g cm}^{-3}$$

$$\rho_s = ?$$

Since mercury in both arm of the U-tube at same level, pressure in both arm will be the same.

$\Rightarrow$ Pressure exerted by water column = pressure exerted by spirit column.

$$P_s = P_w$$

$$\rho_s g h_s = \rho_w g h_w$$

$$\rho_s = \frac{\rho_w h_w}{h_s} = \frac{1 \times 10}{12.5} = 0.80 \text{ cm}^{-3}$$

Specific gravity of spirit $\rho_{rs} = \frac{\text{Density of spirit}}{\text{Density of water}}$

$$= \frac{\rho_s}{\rho_w} = \frac{0.8}{1.0}$$

(2) Two syringes of different cross sections (without needles) filled with water are connected with a tightly fitted rubber tube filled with water. Diameters of the smaller piston and larger piston are 1.0 cm and 3.0 cm respectively.

(a) Find the force exerted on the larger piston when a force of 10 N is applied to the smaller piston.

(b) If the smaller piston is pushed in through 6.0 cm, how much does the larger piston move out.

Solution: (a) Pressure is equal in all direction

$$f_2 = \frac{A_2}{A_1} f_1, \quad f_2 = \frac{(\pi r^2)}{(\pi r^2)} \times f_1 = \frac{\pi \left(\frac{3}{2} \times 10^{-2}\right)^2}{\pi \left(\frac{1}{2} \times 10^{-2}\right)^2} \times 10$$

$$f_2 = 90 \text{ N}$$

~~f = 10 N~~

(b) Since water in the volume is incompressible;

$$V_1 = V_2$$

$$A_1 l_1 = A_2 l_2 = \frac{\pi \left(\frac{1}{2} \times 10^{-2}\right)^2}{\pi \left(\frac{3}{2} \times 10^{-2}\right)^2} \times 6 \times 10^{-2} = 0.67 \times 10^{-2} \text{ m}$$

(3) At a depth of 100 m in an ocean (a) What is the absolute pressure

(b) What is the gauge pressure? (c) Find the force acting on the window of area 20 cm x 20 cm of a submarine at this depth, the interior of which is maintained at sea-level atmospheric pressure.

$$(\rho_{\text{sea water}} = 1.03 \times 10^3 \text{ kg m}^{-3}, g = 10 \text{ m s}^{-2})$$

$$P = P_0 + \rho g h$$

Hydrodynamics

This is ~~the fluid~~ the study of fluid in motion. It finds its application in understanding blood flow in the body, water flow in our home plumbing system, lift of airplane in air, steering and sailing of ships among others.

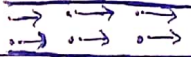
Types of fluid flow

Fluid flow is the motion of fluid subjected to unbalanced forces. Types of fluid flow include:

(1) Laminar and

(2) Turbulent flow

(a) **Laminar flow:** This is a kind of flow where the particles of the fluid move with constant velocity relative to one another. The layers of the particles move parallel to each other and it is characterized by a low Reynold's number (Re)

$$Re \leq 2000$$


(b) **Turbulent flow:** A type of flow where the particles' path crosses each other because they move with non-uniform velocity. The flow is characterized by a high Reynold's number $Re \geq 3000$

Laminar	Turbulent
✓ Regular and Smooth flow	✓ Irregular flow
✓ Constant Velocity	✓ Non-uniform Velocity
✓ $Re \leq 2000$; $Re = \frac{\rho v l}{\eta}$	✓ $Re \geq 3000$
✓ flow at low level velocity	✓ flow at high level velocity
✓ Liquid layers move parallel to each other	✓ Liquid layers move in disorderly manner e.g. oil transportation inside pipelines, sea current, blood flow inside the arteries

Where ρ = density of fluid
 v = Velocity of the liquid
 l = length
 η = Coefficient of viscosity

Equilibrium of bodies in fluid

This is governed by Archimede's principles and principles of floatation

① Archimede's principles: This states that when a body is immersed fully or partially in a fluid, it experiences an upward force that is equal to the weight of the fluid displaced by it.

$$\text{Upward force (upthrust)} = \text{Weight in air} - \text{Weight in water}$$

$$= W_{\text{air}} - \text{Apparent weight}$$

$$= \text{Weight of liquid displaced}$$

$$\boxed{U = W_{\text{liquid displaced}}} = \text{loss in weight}$$

Note:

$$\checkmark \text{ Upward force} = \text{Weight of liquid displaced}$$

$$\checkmark \text{ loss in weight} = \text{real weight} - \text{apparent weight}$$

$$\text{Upward force} = \text{loss in weight}$$

$$\checkmark \text{ Upward force} = W_{\text{liquid displaced}}$$

$$= mg$$

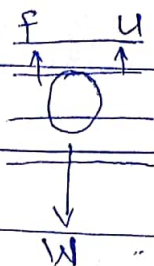
$$= V_L \rho_L g$$

$$\checkmark \text{ Upward force} = \text{Volume of water displaced}$$

Principle of floatation

✓ Floating objects displace its own weight of fluid. or

✓ an object will float when the upthrust exerted upon it by the fluid in which it floats is equal to the weight of the body.



If the liquid is a viscous liquid; it experiences a drag

F if not $F = 0$

for floatation

$$W = U + F$$

$$W = U; \text{ non viscous fluid}$$

$$U = W = W$$

↓ ↓ ↓
Upthrust weight of an object weight of fluid displaced

Application of Principles of floatation

① Hydrometer

② Balloons

③ Boats

④ Ships

Viscosity

Viscosity, factors affecting Viscosity

This is the ability of a fluid to resist flow. There are differences in the ability of fluid to flow e.g rate of flow of honey, Syrup, water, kerosine and petrol. The frictional force (Viscosity, F) in fluids is to the area, A and the Velocity gradient V_g of the liquid layer.

$$F \propto A \times V_g; V_g = \frac{Vel}{L}; L = r$$

$$F \propto \pi r^2 \times \frac{V}{r}$$

$$F = 6\pi\eta r v \quad \dots \text{①}$$

$$F \propto \pi r v$$

η = Coefficient of Viscosity

r = radius of the sphere

v = Velocity of the body

Stoke's Law

Equation ① is called the Stoke's Law

✓ Stoke's law states that the Viscous force (drag force), F acting on a spherical object in a fluid depends on (a) r_{obj} (b) η_{liquid}

③ Thermal Vel. V_0 $F_r = 6\pi\eta r^2 V_0 \quad \dots \text{①}$

$$6\pi\eta r V_0$$

Factors affecting Viscosity

① Intermolecular forces ↑ stickier ↑ ~~the~~ Viscosity

② Shape of molecules tiny molecules $\Rightarrow$ less viscous

③ Temperature $\uparrow$ temp. $\downarrow$ interm. force $\Rightarrow \downarrow$ viscosity

④ Relative Velocity

⑤ pressure

⑥ Impurity

⑦ Surface area of contact = ②

⑧ Concentration

Uses of Viscosity

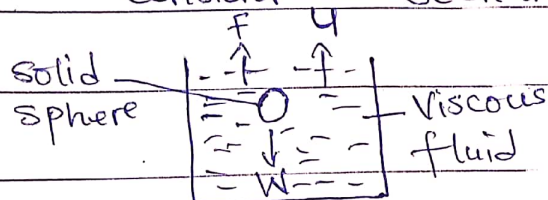
① Lubricants

② Design of Ships and aircraft using Viscous drag

Equation of terminal Velocity

Terminal Velocity is the maximum constant velocity attained by a body when falling through a viscous fluid.

Consider the scenario below:

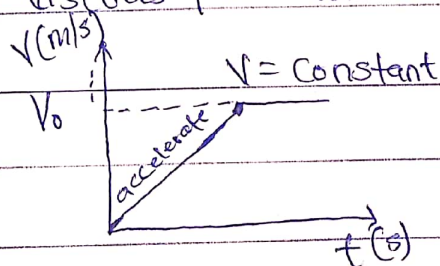


$$W = F + u; a \uparrow$$

$$W = F + u; F = 0, a = 0$$

A graph showing the motion of the sphere $v = \text{constant}$.

in the viscous fluid will look like



$$W = U + F_d$$

$$* U = W_{\text{fluid displaced}} = W_{\text{sphere}}$$

(Archimedes principle)

$$U = mg = \rho_f V g = \frac{4}{3} \pi r^3 \rho_f g \quad \text{--- (1)}$$

$$W = mg = \rho_s V_s g = \frac{4}{3} \pi r^3 \rho_s g \quad \text{--- (2)}$$

$$f = 6 \pi \eta r v \quad \text{--- (3) (Stokes's Law)}$$

at velocity; $v = v_0$ and $W = U + f$

$$\frac{4}{3} \pi r^3 \rho_s g = \frac{4}{3} \pi r^3 \rho_f g + 6 \pi \eta r v_0$$

$$\eta = \frac{2gr^2(\rho_s - \rho_f)}{9v_0} \quad \text{--- (4) another form of Stokes Law}$$

$$v_0 = \frac{2gr^2(\rho_s - \rho_f)}{9\eta} \quad \text{--- (5) Terminal Velocity of the sphere}$$

Determination of Coefficient of Viscosity of a Liquid

- (1) Viscometer
- (2) Poiseuille's method
- (3) Fall measuring graduated cylinder